

InterviewAce Project Charter

Project Title: InterviewAce — AI-Powered Interview Practice Platform

Course: CS 4800

Project Type: Full-Stack AI-Powered Web Application

Repository: github.com/Youcheng9/Job-Interview-Prep-Tool

1. Project Overview

InterviewAce is a full-stack AI-powered interview preparation platform designed to help students and job seekers practice technical and behavioral interview questions with real-time scoring, feedback, and progress tracking. The system combines React frontend interfaces, FastAPI backend services, PostgreSQL persistence, AI/ML scoring using sentence-transformers, deterministic rubric-based evaluation, and AI coaching through Ollama.

2. Project Purpose

The purpose of InterviewAce is to provide users with structured, measurable interview preparation using AI-powered scoring and actionable feedback. The platform improves interview readiness while reducing dependency on expensive coaching services.

3. Business Need

Many interview-preparation platforms provide static question lists without personalized evaluation or measurable feedback. InterviewAce addresses this gap by delivering deterministic rubric-based scoring, AI coaching, answer history tracking, and measurable progress analytics.

4. Project Objectives

- Secure JWT-based authentication
- Role-based interview question banks
- AI-powered scoring and feedback
- AI Coach chat integration
- Responsive React frontend
- PostgreSQL persistence
- Docker deployment and CI/CD

5. Project Scope

In Scope: React frontend, FastAPI APIs, PostgreSQL database, AI scoring engine, Docker deployment, GitHub Actions CI/CD.

Out of Scope: Native mobile apps, video interviews, payment systems, enterprise multi-tenant systems.

6. Architecture Summary

Presentation Layer: React, TypeScript, Tailwind CSS, Recharts.

Application/API Layer: FastAPI, JWT Authentication.

Service Layer: AuthService, QuestionService, ScoringService, HistoryService.

AI/ML Layer: sentence-transformers, cosine similarity scoring, Ollama AI Coach.

Data Layer: PostgreSQL, SQLAlchemy ORM, Alembic.

Infrastructure Layer: Docker, docker-compose, GitHub Actions.

7. Design Pattern

InterviewAce uses the Service Layer Pattern to separate routing logic from business workflows. Service modules coordinate authentication, scoring, feedback generation, and history tracking.

8. Major Use Cases

- Register/Login
- Select Role
- Practice Interview
- Submit Answer
- View Score & Feedback
- AI Coach Chat
- Track Progress
- Manage Question Bank

9. Risks & Mitigation

AI scoring latency → optimized scoring workflow.

Security vulnerabilities → JWT expiration and protected endpoints.

Deployment inconsistencies → standardized Docker setup.

Database migration failures → Alembic migration control.

10. Success Criteria

The project is considered successful if all functional requirements are implemented, AI scoring functions correctly, responsive frontend behavior is maintained, and all major test cases pass successfully.

11. Development Timeline

Weeks 1–3: Backend + Database Setup

Weeks 4–5: AI/ML Pipeline

Weeks 6–7: Frontend Development

Weeks 8–9: Integration & Deployment

12. Stakeholders

Stakeholder	Role
Freeman Yiu	Frontend & AI Systems Lead
Youcheng Taing	Backend & ML Lead
CS 4800 Instructor	Project Sponsor / Evaluator
Job Seekers	Primary End Users

13. Approval

Name	Role
Freeman Yiu	Frontend & AI Systems Lead
Youcheng Taing	Backend & ML Lead
CS 4800 Instructor	Project Sponsor / Evaluator